

Nested FIML Satorra–2000 in Saturated Eta Space

June 28, 2026

Purpose

The question is whether magmaan’s missing-data nested `method = "restriction_map"` statistic has an analytical target, or whether it is only a diagnostic convention. The answer is conditional.

Under the ordinary nested null, where the smaller model is correct in the saturated first-stage eta target, the eta-space statistic is the standard local Satorra restricted-test derivation with the complete-data sample moments replaced by the saturated FIML eta estimator. The finite-sample objects used by magmaan,

$$V = \hat{H}_\eta, \quad \hat{\Gamma}_{\text{mis}} = \hat{H}_\eta^{-1} \hat{J}_\eta \hat{H}_\eta^{-1},$$

are plug-in estimates of the curvature and first-stage sandwich covariance. The estimated curvature or two-stage weight does not add a first-order term at this null, because both fitted residuals are $O_p(N^{-1/2})$. Thus “robust” here means robust to the data distribution and missing-data first-stage covariance under the true nested null; it does not mean robust to structural misspecification of the larger model.

The same derivation does not prove the statistic for the broader null in which the nested restriction holds only at the pseudo-true point while the larger model remains misspecified. In that regime the residual is nonzero, and the limiting law is the Hessian of the two profiled discrepancies, including curvature and estimated-weight derivative channels.

1 Generic eta-space nested result

Let $\hat{\eta}$ be a first-stage estimator of a saturated moment vector η_0 , with

$$Z_N = \sqrt{N}(\hat{\eta} - \eta_0) \Rightarrow Z, \quad Z \sim N(0, \Gamma).$$

Let the larger model M_1 contain η_0 and have local coordinates $\alpha \in \mathbb{R}^r$,

$$\eta_1(\alpha_N) = \eta_0 + N^{-1/2}\Delta\alpha + o_p(N^{-1/2}), \quad \Delta \in \mathbb{R}^{q \times r}.$$

The smaller model $M_0 \subset M_1$ is represented locally by

$$R\alpha = 0, \quad R \in \mathbb{R}^{m \times r}, \quad \text{rank}(R) = m.$$

Assume the local contrast for model M has the quadratic representation

$$T_M = \min_t (Z_N - \Delta_M t)^\top V (Z_N - \Delta_M t) + o_p(1),$$

where V is symmetric positive definite and Δ_M is the local tangent matrix for model M . For the larger model write

$$A = \Delta^\top V \Delta, \quad B = \Delta^\top V \Gamma V \Delta.$$

Proposition 1 (Nested eta-space restriction map). *Under the assumptions above,*

$$T_0 - T_1 \Rightarrow (R\hat{\alpha})^\top C^{-1}(R\hat{\alpha}), \quad \hat{\alpha} = A^{-1}\Delta^\top V Z, \quad C = RA^{-1}R^\top.$$

Equivalently,

$$T_0 - T_1 \Rightarrow \sum_{j=1}^m \lambda_j \chi_{1j}^2, \quad \lambda_j = \text{eig}_j(C^{-1}S),$$

where

$$S = RA^{-1}BA^{-1}R^\top.$$

Proof. The unconstrained minimizer in the larger tangent space is

$$\hat{\alpha}_N = A^{-1}\Delta^\top V Z_N + o_p(1).$$

The smaller model minimizes the same quadratic subject to $R\alpha = 0$. The ordinary equality-constrained quadratic-program identity gives

$$T_0 - T_1 = (R\hat{\alpha}_N)^\top (RA^{-1}R^\top)^{-1}(R\hat{\alpha}_N) + o_p(1).$$

Since

$$\hat{\alpha}_N \Rightarrow A^{-1}\Delta^\top V Z, \quad \text{Var}(\hat{\alpha}) = A^{-1}BA^{-1},$$

the covariance of $R\hat{\alpha}$ is S . The quadratic form $(R\hat{\alpha})^\top C^{-1}(R\hat{\alpha})$ therefore has the weighted chi-square law in (1), by simultaneous whitening of the pair (C, S) . $\square$

Corollary 1 (Normal-theory collapse). *If $\Gamma = V^{-1}$, then $B = A$, $S = C$, and every nonzero eigenvalue in (1) is one. Hence $T_0 - T_1 \Rightarrow \chi_m^2$.*

Proof. Substitution gives $B = \Delta^\top V V^{-1} V \Delta = A$, so

$$S = R A^{-1} A A^{-1} R^\top = R A^{-1} R^\top = C.$$

□

2 FIML specialization

For raw incomplete continuous data, the saturated observed-data normal H1 model has eta vector

$$\eta = (\mu, \text{vech } \Sigma)$$

per independent group or block. Let $\hat{\eta}$ be the saturated EM/FIML estimator. With observed-data normal log likelihood $\ell_i(\eta)$, define

$$H_\eta = -\mathbb{E} \left\{ \frac{\partial^2 \ell_i(\eta_0)}{\partial \eta \partial \eta^\top} \right\}, \quad J_\eta = \mathbb{E} \left[\frac{\partial \ell_i(\eta_0)}{\partial \eta} \frac{\partial \ell_i(\eta_0)}{\partial \eta^\top} \right].$$

The first-stage sandwich covariance is

$$\Gamma_{\text{mis}} = H_\eta^{-1} J_\eta H_\eta^{-1}.$$

The saturated likelihood expansion around $\hat{\eta}$ gives, for local $\eta = \eta_0 + N^{-1/2}h$,

$$2\{\ell_N(\hat{\eta}) - \ell_N(\eta)\} = (Z_N - h)^\top H_\eta (Z_N - h) + o_p(1),$$

so (1) holds with $V = H_\eta$ and $\Gamma = \Gamma_{\text{mis}}$. If the larger SEM has local Jacobian

$$\Delta_1 = \frac{\partial \eta_1(\theta)}{\partial \alpha^\top}$$

after equality-constraint tangent projection, and the smaller SEM is the local restriction $R\alpha = 0$, Proposition 1 gives

$$\begin{aligned} A_1 &= \Delta_1^\top H_\eta \Delta_1, & B_1 &= \Delta_1^\top H_\eta \Gamma_{\text{mis}} H_\eta \Delta_1, \\ C &= R A_1^{-1} R^\top, & S &= R A_1^{-1} B_1 A_1^{-1} R^\top, \end{aligned}$$

and the Satorra–2000 mean scaling

$$\hat{c} = \frac{1}{m} \text{tr}(C^{-1}S).$$

This is exactly the route implemented by magmaan’s FIML nested restriction map: the code constructs V from saturated EM information, uses $\hat{\Gamma}_{\text{mis}}$ from saturated score meat, builds Δ_1 in the same eta layout, obtains R from the H1-to-H0 local restriction, and solves the generalized eigenproblem for the pair (S, C) .

3 Why the estimated metric does not enter at this null

The FIML implementation uses finite-sample plug-in matrices, not fixed population matrices. This does not change the first-order nested null law. Let $\hat{V} = V + o_p(1)$, and let $r_M(\hat{\theta}_M)$ be the fitted eta-space residual for M . If M_0 is true in the saturated eta target, both

$$r_0(\hat{\theta}_0) = O_p(N^{-1/2}), \quad r_1(\hat{\theta}_1) = O_p(N^{-1/2}).$$

Therefore

$$Nr_M(\hat{\theta}_M)^\top (\hat{V} - V) r_M(\hat{\theta}_M) = o_p(1).$$

The same argument covers a smooth two-stage weight $\hat{W} = W + o_p(1)$ under ML2S. Weight-estimation derivative terms are multiplied by the pseudo-true residual. At the true nested null that residual is zero, so they vanish from the first-order reference law.

The finite-sample scaling convention also cancels in the generalized eigenproblem. magmaan stores the saturated information in summed log-likelihood units, $V_N \approx NV$, and the sandwich covariance in $1/N$ units, $\Gamma_N \approx \Gamma/N$. Then

$$A_{1,N} = NA_1, \quad B_{1,N} = NB_1, \quad C_N = N^{-1}C, \quad S_N = N^{-1}S,$$

so the eigenvalues of (S_N, C_N) equal those of (S, C) .

4 ML2S specialization

ML2S uses the same first-stage saturated EM estimator $\hat{\eta}$, then fits the structural model by a moment-quadratic criterion with a stage-two weight W . If W is treated as the local metric in (1), Proposition 1 applies with

$$A_1 = \Delta_1^\top W \Delta_1, \quad B_1 = \Delta_1^\top W \Gamma_{\text{mis}} W \Delta_1.$$

For the normal-theory two-stage weight $W = \Gamma_{\text{NT}}^{-1}$ and normal complete-data first-stage law, Corollary 1 again yields unit eigenvalues. For empirical robust ML2S, Γ_{mis} is the saturated EM sandwich and the eigenvalues are the robust nested reference spectrum.

When the stage-two weight is estimated from the same first-stage eta, its derivative is still asymptotically irrelevant at the true nested null by the argument in Section 3. It is not irrelevant when the nested restriction is tested inside a misspecified larger model.

5 Boundary of the derivation

The derivation above assumes the smaller model contains the first-stage target:

$$\eta_0 \in M_0 \subset M_1.$$

This is the usual calibration null for a nested difference test and the regime in which size is defined when the data-generating model satisfies the smaller model. It is data-robust in the Satorra–Bentler/Yuan–Bentler sense: the sandwich covariance may differ from the normal-theory covariance because the observed data are nonnormal or incomplete, but the model restriction being tested is still evaluated under a correctly specified smaller model.

A different null is possible: the larger and smaller profiled criteria have the same pseudo-true optimum because the restriction is true inside a misspecified larger model, but the common fitted residual r_* is nonzero. Then the first derivative of the profiled difference still cancels, but the second derivative is not the residual-projector covariance above. It is the profile Hessian difference

$$Z^\top(Q_0 - Q_1)Z,$$

where Q_M contains both model curvature terms and, when the metric is estimated, derivatives of the metric map. This is the regime isolated in the relative-fit and nested-profile note. The eta-space restriction map is recovered from that profile-Hessian law only in special cases: fixed metric, zero pseudo-true residual, or cancellations that make the profile Hessian equal the ordinary Satorra projector.

6 Implications for the lavaan parity gap

The missing-data divergence from `lavTestLRT(method = "satorra.2000")` is therefore not, by itself, an oracle defect or a magmaan defect. It says that lavaan and magmaan are using different missing-data conventions for the difference-test scaling quantity. The eta-space route has a first-principles null derivation for saturated FIML and ML2S under the true nested null. It is the robust method under study.

Two additional claims would require separate work.

1. Lavaan compatibility: derive and expose lavaan’s missing-data finite-sample convention as a separate route, then gate that route against lavaan.

2. Misspecified nested robustness: implement or approximate the full profile-Hessian law for $r_* \neq 0$, including estimated-metric derivative terms, then calibrate it by simulation.

References

- [1] Satorra, A. (1989). Alternative test criteria in covariance structure analysis: A unified approach. *Psychometrika*, 54(1), 131–151. doi:10.1007/BF02294453.
- [2] Satorra, A. (2000). Scaled and adjusted restricted tests in multi-sample analysis of moment structures. In R. D. H. Heijmans, D. S. G. Pollock, & A. Satorra (Eds.), *Innovations in Multivariate Statistical Analysis* (pp. 233–247). Kluwer. doi:10.1007/978-1-4615-4603-0_17.
- [3] Satorra, A., & Bentler, P. M. (2001). A scaled difference chi-square test statistic for moment structure analysis. *Psychometrika*, 66(4), 507–514. doi:10.1007/BF02296192.
- [4] Savalei, V., & Bentler, P. M. (2009). A two-stage approach to missing data: Theory and application to auxiliary variables. *Structural Equation Modeling*, 16(3), 477–497. doi:10.1080/10705510903008238.
- [5] White, H. (1982). Maximum likelihood estimation of misspecified models. *Econometrica*, 50(1), 1–25. doi:10.2307/1912526.
- [6] Yuan, K.-H., & Bentler, P. M. (2000). Three likelihood-based methods for mean and covariance structure analysis with nonnormal missing data. *Sociological Methodology*, 30, 165–200. doi:10.1111/0081-1750.00078.